Status	Finished
Started	Friday, 16 October 2026, 11:27 AM
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Duration	3 mins 17 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O👎 ✓
d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The first element is returned
c. The search is considered unsuccessful ✓
d. The last element is returned

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list contains duplicate elements
c. When the list is sorted ✓
d. When the list is very small

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It does not work with negative numbers

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(n log n)
- c. O(1)
- d. O(log n) ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Hash search
- c. Interpolation search
- d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 100

c. 89 and 94

d. 90 and 99 ✓

Question **8**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The lower middle element is chosen as the middle element ✗

b. The search stops

c. The middle element is chosen randomly

d. The higher middle element is chosen as the middle element

Question **9**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The search ends successfully ✓

c. The list is sorted

d. The search continues in the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the list is sorted
- c. When the search algorithm finishes
- d. When the list contains duplicate elements

Question **11**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search continues until the end of the list
- c. The search backtracks to find duplicate elements
- d. The search starts over from the beginning

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Sorting elements in a list
- b. Deleting elements from a list
- c. Determining whether an element is present in a list ✓
- d. Inserting elements into a list

Question **13**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Linear search ✓
- c. Binary search
- d. Jump search

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(n log n)
c. O(log n)
d. O(1) ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Linear search ✓
c. Hash search
d. Both (1) & (3)